

C5.3 Equilibria

Summary

In a reaction, when the rate of the forward reaction equals the rate of the backwards reaction, the reaction in a closed system is said to be in equilibrium.

Underlying knowledge and understanding

Learners will be familiar with representing chemical reactions using formulae and using equations.

Common misconceptions

Learners often do not recognise that when a dynamic equilibrium is set up in a reaction the concentration of the reactants and products remain constant. They think that the concentrations of all substances are equal. Learners also sometimes perceive a dynamic equilibrium as two reactions.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM5.3i	arithmetic computation, ratio when measuring rates of reaction	M1a, M1c
CM5.3ii	drawing and interpreting appropriate graphs from data to determine rate of reaction	M4b, M4c
CM5.3iii	determining gradients of graphs as a measure of rate of change to determine rate	M4d, M4e
CM5.3iv	proportionality when comparing factors affecting rate of reaction	M1c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C5.3a	recall that some reactions may be reversed by altering the reaction conditions	M1a, M4b, M4c		
C5.3b	recall that dynamic equilibrium occurs in a closed system when the rates of forward and reverse reactions are equal	M4b, M4c		
C5.3c	predict the effect of changing reaction conditions on equilibrium position and suggest appropriate conditions to produce as much of a particular product as possible	M1a, M4d, M4e, M1c	WS1.2a, WS1.2b, WS1.2c, WS1.4c, WS2a, WS2b	